



RIVER VALLEY HIGH SCHOOL
2020 JC2 Preliminary Examination
Higher 2

NAME	
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CLASS					
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INDEX NUMBER		
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MATHEMATICS

9758/01

Paper 1

17 September 2020

3 hours

Candidates answer on the Question Paper
 Additional Materials: List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Write your answers in the spaces provided in the question paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved graphing calculator is expected, where appropriate.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 100.

For examiner's use only	
Question number	Mark
Total	

Calculator Model:

1. Let $f(x) = \ln(\cos x)$. Using the standard series, find the Maclaurin series for $f(x)$, up to and including the term in x^4 . Hence, find the Maclaurin series of $\tan x$ up to and including the term in x^3 . [6]

2. (a) Sketch on the same axes, the graphs of $y = 1 - x - \frac{1}{x-1}$ and $y = |ax - a|$, where $-1 < a < 1$. State clearly the equation of any asymptotes and coordinates of any turning points and axial intercepts. [5]
- (b) Suppose that a can be any real number, state the largest possible set of solution for the inequality $1 - x - \frac{1}{x-1} > |ax - a|$, and the corresponding range of values of a . [2]

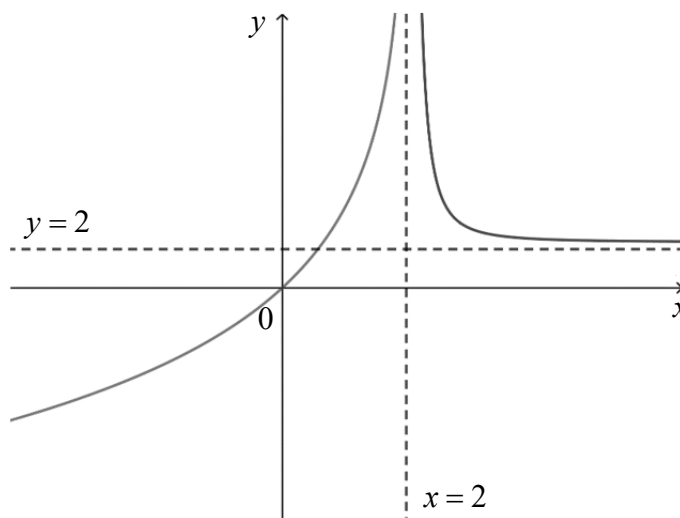
3. The vectors $\mathbf{u}$ and $\mathbf{v}$ are unit vectors. The angle θ between $\mathbf{u}$ and $\mathbf{v}$ is an acute angle.
 - (i) Show that $(\mathbf{u} \cdot \mathbf{v} + |\mathbf{u} \times \mathbf{v}|)^2 = 1 + \sin 2\theta$. [3]
 - (ii) The vector $\mathbf{w}$ is such that $\mathbf{u} + \mathbf{w} = k\mathbf{v}$, where k is a constant in terms of θ . By considering $(\mathbf{u} + \mathbf{w}) \cdot \mathbf{v}$ or otherwise, find the value of k if $\mathbf{w}$ is also a unit vector. [3]
 - (iii) Deduce the range of possible values of k . [1]

4. (i) State a pair of transformations that will transform the graph of $y = f(x)$ onto the graph of $y = f(2x) - 1$. [2]

It is given that

$$f(x) = \begin{cases} x^2 & , x \leq 2 \\ 2 - \frac{1}{x-2} & , x > 2 \end{cases}$$

- (ii) Sketch on separate diagrams, the graphs of $y = f(x)$ and $y = f(2x) - 1$. [4]
- (iii) It is proposed that the following graph is the graph of $y = f'(x)$.



Give two reasons to explain why the proposed graph is not suitable to be the graph of $y = f'(x)$. [2]

5. (a) The curve C has parametric equations

$$x = \cos t, \quad y = \sec t + \sin t,$$

where $0 < t < \frac{\pi}{2}$.

Find $\int y \, dx$ in terms of x . [5]

- (b) Find the exact value of $\int_{-\sqrt{2}}^1 |x^3 + 1| \, dx$. [3]

6. An *astroid* is a curve with parametric equations

$$x = a \cos^3 t, \quad y = a \sin^3 t.$$

- (i) Sketch an astroid with $a = 2$ and $0 \leq t \leq \frac{\pi}{2}$, indicating clearly the coordinates of all axial intercepts. [2]

For the rest of the question, take $a = 1$.

- (ii) Find the equation of the tangent to the curve at the point P where $t = p$, simplifying your answer. [4]
- (iii) The tangent to the curve at P intersects the x - and y -axes at the points Q and R respectively. Find the cartesian equation for the locus of the midpoint of QR . [3]

7. (i) Given that $f(r) = \ln(3r+1)$, show that $f(r) - f(r-1) = \ln\left(\frac{3r+1}{3r-2}\right)$. [1]

- (ii) Hence evaluate $\sum_{r=1}^n \ln \sqrt{\frac{3r+1}{3r-2}}$. [4]

- (iii) Show that $\sum_{r=4}^{n-5} \ln \sqrt{\frac{3r+16}{3r+13}} = \sum_{r=9}^n \ln \sqrt{\frac{3r+1}{3r-2}}$. Hence find $\sum_{r=4}^{n-5} \ln \sqrt{\frac{3r+16}{3r+13}}$ in terms of n . [4]

8. The curves C_1 and C_2 are given by the equations $4x^2 + y^2 = 16$ and $y = x^2 - x - 2$ respectively. The region R is bounded by C_1 and C_2 , and $x \geq 0$.

- (a) (i) Using the substitution $x = 2 \sin u$, find $\int \sqrt{16 - 4x^2} \, dx$. [4]
- (ii) Hence find the exact area of R . [4]
- (b) The region S is bounded by C_2 , the lines $x = -1$ and $x = 1$, and the x -axis. Find the volume generated when S is rotated 2π radians about the x -axis. [2]

9. (a) Solve the simultaneous equations

$$z - 2w^* = i,$$

$$iz - w = i. \quad [5]$$

- (b) u and v are complex numbers such that $u = -1 - \sqrt{3}i$, $|v| = \sqrt{2}$ and $\arg(v) = \frac{7\pi}{12}$.

- (i) Find the modulus and argument of u . [2]
- (ii) Express vu^* in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$. [2]
- (iii) Find the least positive integer n such that $(vu^*)^n$ is purely imaginary. [2]

10. Sales agent A started work on 1st June 2020. He plans to acquire 2 clients in his first month of work. For subsequent months, the number of clients he plans to acquire per month is 3 more than that in the previous month.

(i) Find the number of clients agent A would acquire in June 2021. [2]

(ii) Find the number of months needed for agent A to acquire at least 500 clients in total. [3]

Sales agent B has a total 500 clients on 1st June 2020. Unlike agent A , he acquires 3 new clients every month. However, due to strong competition from other sales agents, agent B loses 1% of his clientele at the end of each month when the clients choose not to continue with his service from the following month.

(iii) Show that the clientele size of agent B at the end of n^{th} month is given by

$$297 + 203(0.99)^n. \quad [4]$$

(iv) The clientele size of agent B will eventually stabilise. State the value of this clientele size. [1]

(v) Agent B wishes to keep a clientele size of at least 350. Find the maximum percentage loss of clients he can have at the end of each month in order to meet this target. [3]

11. (a) Sketch the graph of $y = (\sin x + 1.1)^{-0.5}$ for $0 \leq x \leq 2\pi$ and state the coordinates of the maximum and minimum points of the curve. [3]

- (b) A comet is an icy celestial body that usually orbits the sun in the *Kuiper Belt* – a region of the solar system beyond the planet Neptune. Some comets have highly elliptical orbits that bring them into the inner solar system for short periods of time.

For a particular comet, the distance between the comet and the sun, r astronomical units (AU), at time t years after it is first observed, is described by the equation

$$r = 24 \left(\sin \left(\frac{t}{76} \right) + 1.1 \right)^{0.5}.$$

As the comet approaches the sun, it will gain kinetic energy but lose an equivalent amount of gravitational potential energy, such that the total amount of energy remains constant. This gives the equation

$$E = \frac{1}{2}mv^2 - \frac{GMm}{r},$$

where E is the total energy in the system, m is the mass of the comet, G is the universal gravitational constant, M is the mass of the sun and v is the speed of the comet. The constants E , m , G and M are all positive.

- (i) Show that $v^2 = \frac{GM}{12 \left(\sin \left(\frac{t}{76} \right) + 1.1 \right)^{0.5}} + \frac{2E}{m}$. [1]

- (ii) By differentiation and using the result in part (a), find the maximum speed of the comet in terms of the constants E , m , G and M . [5]

The luminance L of an object is a measure of how much light it is emitting. As the comet leaves the inner solar system, it reflects less light from the sun and becomes less luminous.

- (iii) It is known that, when first observed, the luminance of the comet would decrease by 2 units for each astronomical unit it travelled away from the sun. Using the relationship between r and t , find the rate at which the luminance of the comet is decreasing when it is first observed. [3]

END OF PAPER